1. Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms (Point 5 in the list)

This is the core ability required to solve this problem. Students need to follow basic geometric rules and the definition of a scale to calculate area.

- **Specific Analysis:** Students must understand and apply the rule of the scale: each grid square represents an area of 0.5 m × 0.5 m, or 0.25 square meters. They need to count the number of squares in the relevant areas and then apply the area calculation algorithm (number of squares × 0.25 m^2) to find the total area. This is a direct application of established mathematical rules and algorithms (area calculation).

2. Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation (Point 12 in the list)

Calculating the total area for the required regions involves multiple steps.

- **Specific Analysis:**
  1. Determine the calculation strategy: either calculate the total area of the whole shop and subtract the service and counter areas, or just sum the areas of the entrance and seating areas.
  2. Count the number of squares in the relevant regions (e.g., seating area and entrance area).
  3. Calculate the actual area of a single square (0.5 m × 0.5 m = 0.25 m^2).
  4. Multiply the total number of squares by the area of a single square.
     This is a clear multi-step procedure.

3. Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations (Point 7 in the list)

This task requires students to manipulate multiple types of information.

- **Specific Analysis:** Students need to extract information from the graphical representation (floor plan) by counting squares, manipulate numbers (applying the scale, addition, and multiplication), and present the result as a final area value. This involves switching between different representations and manipulating data.

4. Articulate a solution, showing and/or summarising and presenting intermediate mathematical results (Point 8 in the list)

The question explicitly asks to "Show your work".

- **Specific Analysis:** Students must provide not just the final answer, but also clearly display how they counted the squares and performed the area calculation. This emphasizes the ability to communicate and document mathematical reasoning.

1. 利用基于定义、规则和形式系统的构建，包括运用熟悉的算法 (列表中的第 5 点)

这是解决此问题的核心能力。学生需要遵循基本的几何规则和比例尺定义来计算面积。

- **具体分析：** 学生必须理解并应用比例尺的规则：每个网格代表 0.5 m × 0.5 m 的面积，即 0.25 平方米 (p. 1)。他们需要通过清点相关区域的网格数量，然后应用面积计算算法（网格数 × 0.25 m^2）来得出总面积。这是对既定数学规则和算法的直接应用。

2. 利用多步骤程序得出数学解、结论或概括 (列表中的第 12 点)

计算所需区域的总面积需要多个步骤。

- **具体分析：**
  1. 确定计算策略：可以计算整个商店的总面积，然后减去服务区和柜台的面积；或者只计算入口区域和座位区域的面积总和。
  2. 清点相关区域（例如，座位区和入口区）的网格数量。
  3. 计算单个网格的实际面积（0.5 m × 0.5 m = 0.25 m^2）。
  4. 将网格总数乘以单个网格的面积。
     这是一个清晰的多步骤过程。

3. 操作数字、图形和统计数据及信息、代数表达式和方程式以及几何表示 (列表中的第 7 点)

该任务要求学生操作多种类型的信息。

- **具体分析：** 学生需要从图形表示（平面图）中提取信息（清点网格），操作数字（应用比例尺、加法和乘法），并将结果表示为最终的面积数值。这涉及在不同表示形式之间切换和操作数据。

4. 阐明解决方案，展示和/或总结并呈现中间数学结果 (列表中的第 8 点)

题目明确要求“Show your work”（展示你的工作）(p. 1)。

- **具体分析：** 学生不仅要提供最终答案，还需要清晰地展示他们是如何清点网格以及如何进行面积计算的。这强调了沟通和记录数学推理的能力。